

Iris EDA and Classification Project Documentation

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# Project Documentation
## Iris EDA and Classification Project

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# 1. Learning Notes

## Topics Learned

### Exploratory Data Analysis (EDA)
Learned how to analyze datasets before applying machine learning models.

Covered:
- Dataset inspection
- Missing value checking
- Correlation analysis
- Histograms
- Boxplots
- Pairplots
- Scatterplots
- Violin plots

### Correlation Analysis
Learned how to identify highly correlated features using heatmaps and correlation matrices.

Important finding:
- PetalLengthCm and PetalWidthCm showed strong positive correlation.

### Feature Engineering
Learned:
- Label Encoding
- Feature Scaling using StandardScaler
- Removing unnecessary columns such as Id

### Machine Learning Algorithms
Implemented:
- Logistic Regression
- KNN
- Decision Tree
- SVM

### Model Evaluation
Learned evaluation techniques:
- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve
- AUC Score

### Dimensionality Reduction
Learned how to use UMAP for 3D visualization of high-dimensional data.

### JSON Output Handling
Learned how to:
- Convert Python dictionaries into JSON format
- Store experiment outputs in structured JSON files

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# 2. Experiment Logs

## Experiment 1 – Dataset Loading
### Objective
Load the Iris dataset into Google Colab.

### Result
Successfully loaded dataset using:
pd.read_csv('/content/Iris.csv')

### Status
Success

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## Experiment 2 – Data Inspection
### Objective
Understand dataset structure and statistics.

### Actions
Used:
- head()
- info()
- describe()

### Findings
- Dataset contains 150 rows
- No major datatype issues
- Numerical features available for ML

### Status
Success

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## Experiment 3 – Missing Value Analysis
### Objective
Check for null or missing values.

### Findings
No missing values found.

### Status
Success

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## Experiment 4 – Correlation Analysis
### Objective
Identify relationships between features.

### Findings
Strong correlation:
- PetalLengthCm ↔ PetalWidthCm

### Status
Success

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## Experiment 5 – Data Visualization
### Objective
Visualize feature distributions and class separations.

### Visualizations Used
- Histograms
- Pairplots
- Scatterplots
- Violin plots
- Heatmaps

### Findings
- Setosa class clearly separable
- Virginica and Versicolor overlap slightly

### Status
Success

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## Experiment 6 – Feature Scaling
### Objective
Normalize feature values before training models.

### Method
StandardScaler

### Status
Success

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## Experiment 7 – Machine Learning Models
### Objective
Train multiple classification algorithms.

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### Models Used
- Logistic Regression
- KNN
- Decision Tree
- SVM

### Findings
SVM achieved highest performance.

### Status
Success

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## Experiment 8 – ROC Curve and AUC
### Objective
Evaluate classification quality.

### Findings
- High AUC scores across all models
- SVM achieved best class separation

### Status
Success

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## Experiment 9 – 3D UMAP Visualization
### Objective
Visualize dataset in 3D reduced space.

### Findings
Distinct clusters observed for flower species.

### Status
Success

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## Experiment 10 – JSON Export
### Objective
Store all outputs in structured JSON format.

### Stored Data
- Dataset summary
- EDA statistics
- Feature transformations
- Predictions
- Metrics

### Status
Success

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# 3. Outputs

## Generated Outputs

### Visual Outputs
- Correlation Heatmap
- Histograms
- Pairplots
- Boxplots
- Scatterplots
- Violin plots
- ROC Curves
- 3D UMAP Graph

### Machine Learning Outputs
- Accuracy scores
- Confusion matrices
- Classification reports
- ROC-AUC scores

### Files Generated
- iris_structured_results.json
- iris_project.ipynb

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# 4. Issue Log

## Issue 1 – Correlation Error
### Problem
Species column caused issues during correlation calculation.

### Solution
Used:
df.corr(numeric_only=True)

### Status
Resolved

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## Issue 2 – ROC Curve for Multi-Class Dataset
### Problem
ROC curve initially failed because Iris has 3 classes.

### Solution
Used:
- OneVsRestClassifier
- label_binarize()

### Status
Resolved

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## Issue 3 – SVM Probability Error
### Problem
SVM did not support predict_proba() by default.

### Solution
Used:
SVC(probability=True)

### Status
Resolved

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## Issue 4 – JSON Serialization Errors
### Problem
NumPy arrays could not be directly stored in JSON.

### Solution
Converted arrays using:
.tolist()

### Status
Resolved

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# 5. What Worked

- Data preprocessing pipeline
- Feature scaling
- Multiple ML model training
- ROC-AUC evaluation
- JSON export system
- 3D UMAP visualization
- Correlation analysis
- Model comparison

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# 6. What Failed

- Initial ROC implementation for multiclass classification
- JSON export of NumPy arrays without conversion
- Correlation calculation including categorical columns

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# 7. What Is Not Understood

- Internal mathematical working of SVM kernels
- Advanced ROC-AUC theory for multiclass problems

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- UMAP optimization parameters
- Hyperparameter tuning strategies
- Advanced feature engineering techniques

8. Final Conclusion

This project successfully implemented a complete machine learning workflow using the Iris dataset. The project